

30th Jan

$$TTQ = 0$$

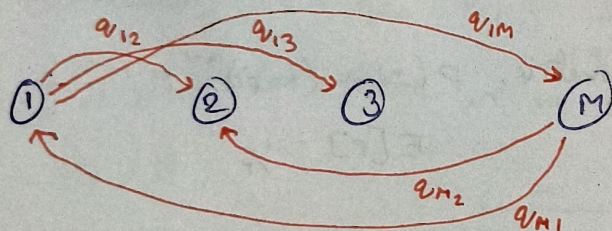
$$S = \{1, 2, \dots, M\}$$

$$Q = \begin{bmatrix} q_{11} & q_{12} & \dots & q_{1M} \\ \vdots & \vdots & \ddots & \vdots \\ q_{M1} & q_{M2} & \dots & q_{MM} \end{bmatrix}$$

→ exponential weights

q_{12} → instate 1 and going to state 2

Transition diagram for CTMC



→ Instead of prob. like DTMC, we denote rates

$$\sum_{j \neq i} q_{ij} = \text{rate of leaving state } i = -q_{ii}$$

$$\pi_i \sum_{j \neq i} q_{ij} = \text{avg. rate of leaving state } i \quad (\text{need not b/w } 0 \text{ and } 1)$$

$$\sum_{j \neq i} \pi_j q_{ji} = \text{avg. rate of entering } i$$

equal

$$[\pi_1 \dots \pi_M] \begin{bmatrix} q_{11} & \dots & q_{1M} \\ \vdots & \ddots & \vdots \\ q_{M1} & \dots & q_{MM} \end{bmatrix}$$

$$\pi Q = 0$$

$$\pi_i q_{ii}$$

$$\pi_i q_{ii} + \sum_{j \neq i} \pi_j q_{ji} = 0$$

π is the stationary distribution

n is large

$$P(X_n = j / X_0 = i) = \pi_j$$

↓
doesn't matter if it is i or k as after a long time is equal as stationary distribution

⇒ Rate at which you enter a state i is equal to rate at which you leave state i .

Subsets of state

$$M \subseteq S$$

$$\pi_i \sum_{\substack{j \neq i \\ j \in M}} q_{ij} = \sum_{\substack{j \neq i \\ j \in M}} \pi_j q_{ji} \rightarrow \text{Partial Balance eq}^n$$

if holds
for all i and all subsets

Global Balance Equation

$$\sum_{j \neq i} \pi_j q_{ji} = \pi_i \sum_{j \neq i} q_{ij} \Rightarrow \pi Q = 0$$

check for all $i \rightarrow 1$ to M
which equals to $\pi Q = 0$

$$\pi_i q_{ik} = \pi_k q_{ki} \rightarrow \text{Detailed Balance eq}^n$$

$$\pi_i q_{i1} + \pi_i q_{i2} + \dots + \pi_i q_{iM} = \pi_1 q_{1i} + \pi_2 q_{2i} + \pi_3 q_{3i} + \dots + \pi_M q_{Mi}$$

↳ see if it's a valid solution and check for all i , if to see if global balance eqⁿ holds

→ for Global we need ^{to check} eqⁿ for all i

$M/M/1/1$

Arrival process
 $PP(\lambda)$

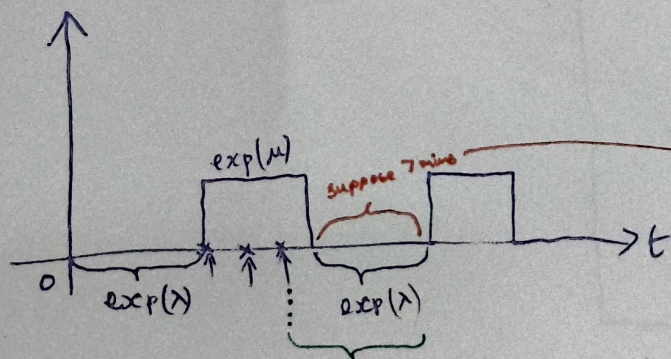
Service time

No. of jobs in the system at a time

No. of servers

$$\{x(t), t \geq 0\} \quad \bullet \quad x(t) = \begin{cases} 1 & \text{if server is busy} \\ 0 & \text{if server is empty} \end{cases}$$

$\mu \rightarrow$ the server samples according to exp. distribution the time taken to do the same k units of work.



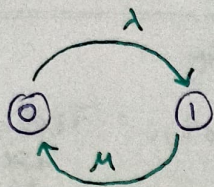
• arrivals when server is busy are lost

$$P(x(t+\tau)=1/x(t)=0)$$

• It doesn't matter when the previous guy arrived, when you have to calculate after time t , the previous info. doesn't matter

exp starts here (the last arrival)
but memory less property so
taken when server became idle (think!)

$$Q = \begin{matrix} & 0 & 1 \\ \begin{matrix} 0 \\ 1 \end{matrix} & \begin{bmatrix} -\lambda & \lambda \\ \mu & -\mu \end{bmatrix} \end{matrix} \rightarrow Q\text{-matrix for } M/M/1/1$$



$$\pi_0 = \frac{\mu}{\lambda + \mu}$$

$$\pi_1 = \frac{\lambda}{\lambda + \mu}$$

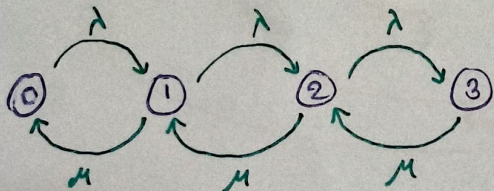
Time avg of a sample path = $\frac{1}{t} \int_{t=0}^t x(t) dt$ $\rightarrow$ similar to law of large nos.

$$E[x(\infty)] = \pi_1$$

(Ergodic processes)
has same time avg
& avg. value at infinite time

M/M/1/∞

$N(t)$ = No. of jobs in the queue at time t



$$\pi(i)(\lambda + \mu) = \pi_{i-1}(\lambda) + \pi_{i+1}(\mu)$$

$$Q = \begin{matrix} & 0 & 1 & 2 & 3 & \dots \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} -\lambda & \lambda & 0 & 0 & \dots \\ \mu & -(\lambda + \mu) & \lambda & 0 & \dots \\ 0 & \mu & -(\lambda + \mu) & \lambda & \dots \\ 0 & 0 & \mu & -(\lambda + \mu) & \dots \end{bmatrix} \end{matrix}$$

~~etc~~

$$\lambda \pi_0 = \mu \pi_1$$

$$\pi_1 = \left(\frac{\lambda}{\mu}\right) \pi_0 = \rho \pi_0$$

$$\pi_2 = \rho^2 \pi_0$$

Using detailed balance eqⁿ,

$$\begin{aligned} \pi_{i+1} &= \rho \pi_i \\ &= \rho^{i+1} \pi_0 \end{aligned}$$

$$\sum \pi_i = 1$$

$$\pi_0 = 1 - \rho$$

ρ : utilisation factor